

TrES-1b

R-1885

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_150713 · created 2026-08-03T12:20:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457216.7983 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.052
Transit depth (Rp/Rs)^2	3.32 +/- 1.91 [%]
Orbital Inclination (inc)	87.56 +/- 1.43
Airmass coefficient 1 (a1)	21.58 +/- 14.96
Airmass coefficient 2 (a2)	-2.94 +/- 1.1
Scatter in the residuals of the lightcurve fit is	68.0 %
Best Comparison Star	None
Optimal Aperture	2.21
Optimal Annulus	7.55
Transit Duration (day)	0.096 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.052 vs 0.1358 (deviation 0.84σ)
Duration fit vs published	0.096 d vs 0.1042 d (deviation 0.52σ)
Mid-time O–C	-2.8 min
Depth SNR	3.3
Residual scatter	68.0 %

Light curve

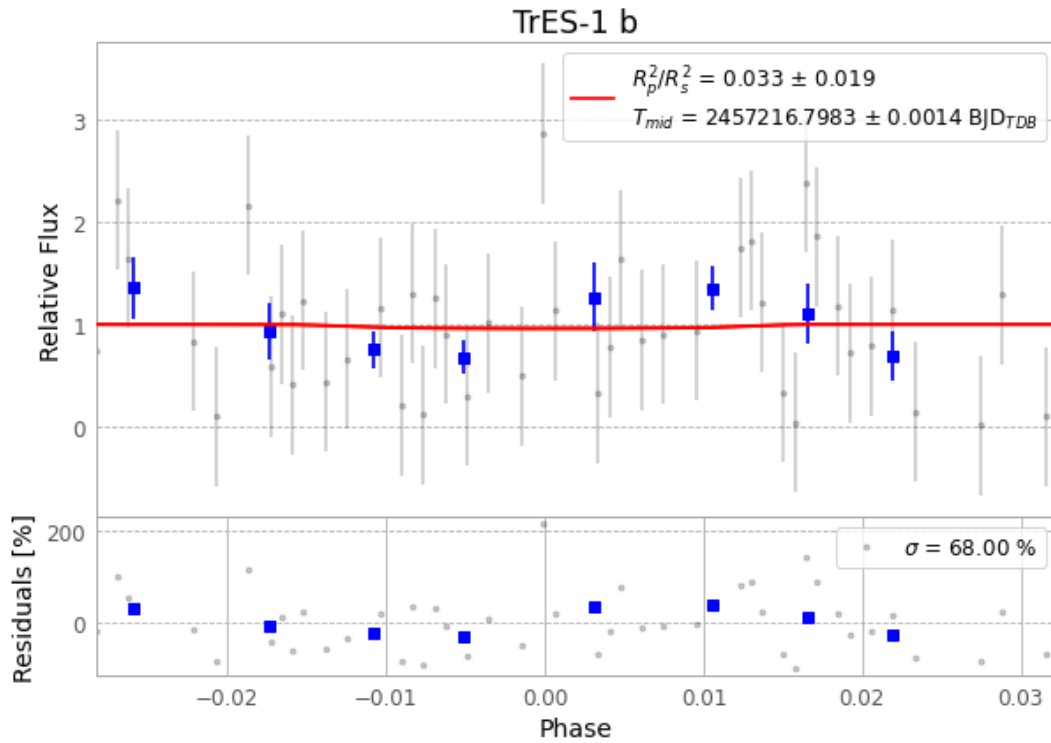


Figure 1 — FinalLightCurve_TrES-1 b_2015-07-12.png (SHA-256 dbc6e1561bc82d7d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 435], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 75e00b06167ad6d3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), TRES-1150713050012.FITS → TRES-1150713092712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-150713.log (710a155fb5c5...), FinalLightCurve_TrES-1 b_2015-07-12.png (dbc6e1561bc8...), FinalLightCurve_TrES-1 b_2015-07-12.csv (53643d1da32e...)

Compute resources

Wall time 250.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:20Z.